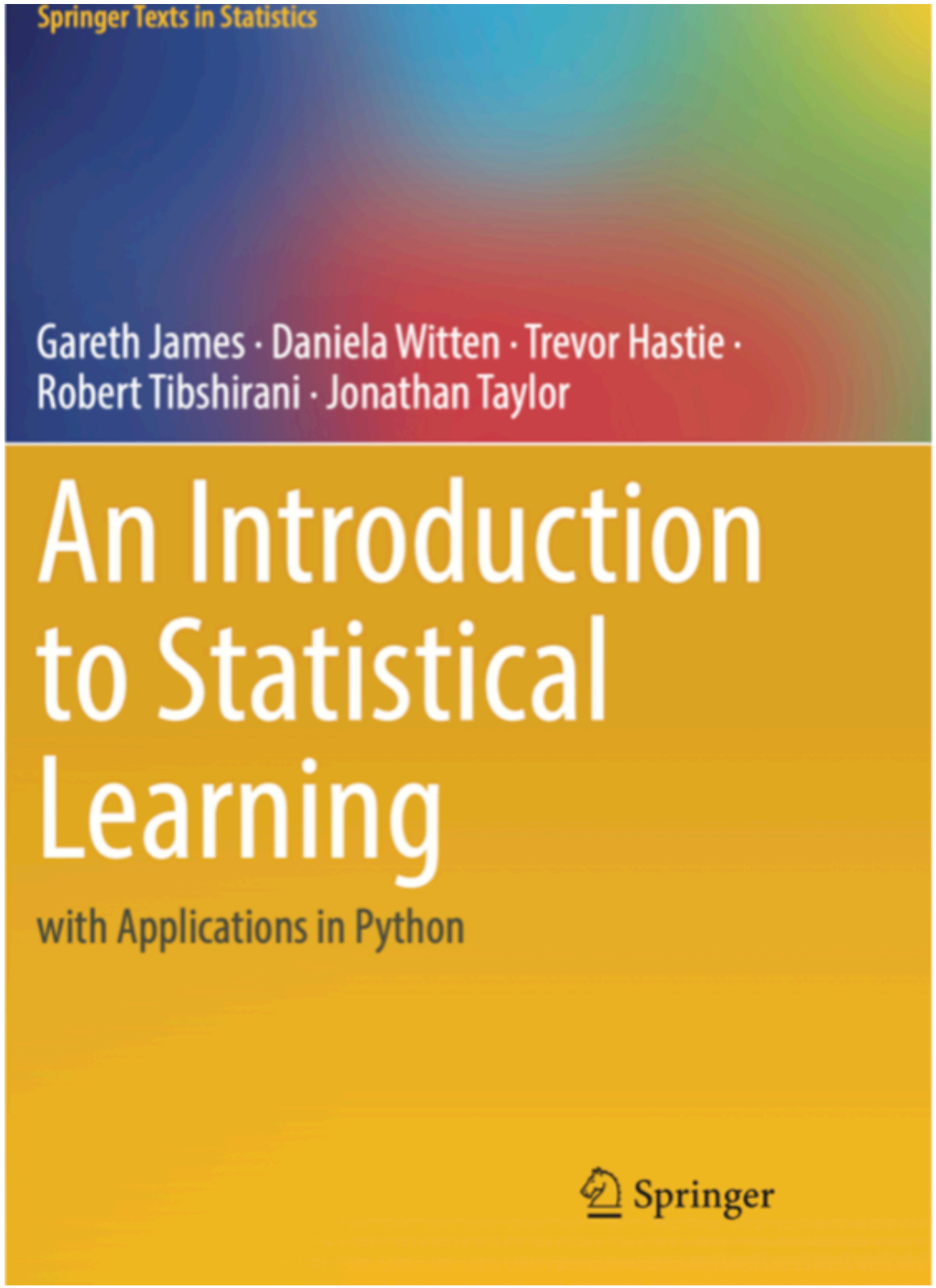
4.4

Generative Models for Classification



https://www.statlearning.com/

Discriminant Analysis

Here the approach is to model the distribution of X in each of the classes separately, and then use *Bayes theorem* to flip things around and obtain $\Pr(Y|X)$.

When we use normal (Gaussian) distributions for each class, this leads to linear or quadratic discriminant analysis.

However, this approach is quite general, and other distributions can be used as well. We will focus on normal distributions.

1701-1761 (aged 59), London, England

The Reverend

Thomas Bayes



Portrait purportedly of Bayes used in a 1936 book,[1] but it is doubtful whether the portrait is actually of him.[2] No earlier portrait or claimed portrait survives.

Alma mater	University of Edinburgh
Known for	Bayesian statistics Bayes' theorem Conditional probability Inverse probability Bayes prior Bayes factor Bayesian inference See full list
	Scientific career
Fields	Probability
	Signature
	

https://en.wikipedia.org/wiki/Thomas_Bayes

Bayes theorem for classification

Thomas Bayes was a famous mathematician whose name represents a big subfield of statistical and probabilistic modeling. Here we focus on a simple result, known as Bayes theorem:

$$\Pr(Y = k|X = x) = \frac{\Pr(X = x|Y = k) \cdot \Pr(Y = k)}{\Pr(X = x)}$$

One writes this slightly differently for discriminant analysis:

$p_k(x)=$ $$\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)},$$ where

Bayes Theorem

- $f_k(x) = \Pr(X = x|Y = k)$ is the *density* for X in class k. Here we will use normal densities for these, separately in each class.
- $\pi_k = \Pr(Y = k)$ is the marginal or *prior* probability for class k.

Law of total probility

Proof

1. The events $\{Y = 1\}, \{Y = 2\}, \ldots, \{Y = K\}$ form a **partition** of the sample space: exactly one of them happens.
2. For any event about X, in particular the event $\{X = x\}$, we can decompose it along that partition:

$$\Pr(X = x) = \sum_{k=1}^{K} \Pr(X = x, Y = k).$$

   This is just "split the probability according to which class occurred."
3. Apply the product rule to each term:

$$\Pr(X = x, Y = k) = \Pr(X = x \mid Y = k) \Pr(Y = k).$$

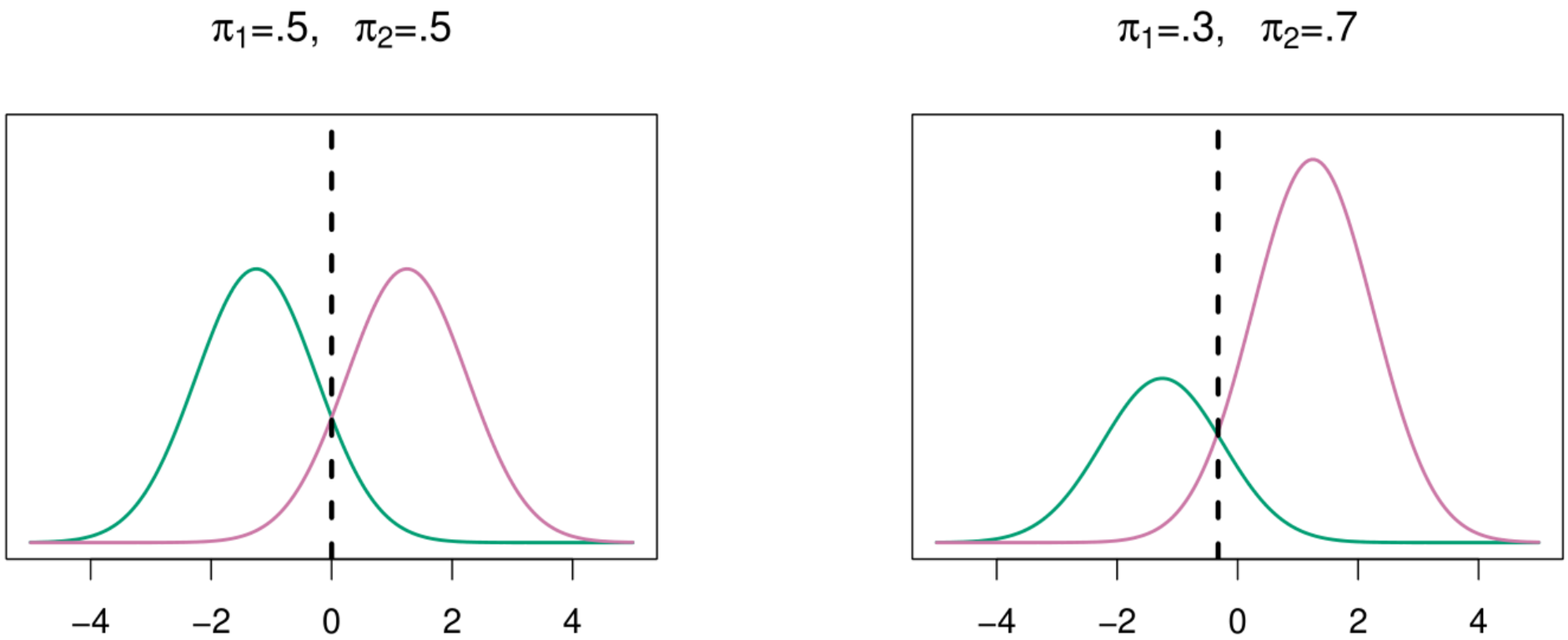
4. Substitute into the sum:

$$\Pr(X = x) = \sum_{k=1}^{K} \Pr(X = x \mid Y = k) \Pr(Y = k).$$

5. Using the snapshot notation $\Pr(X = x \mid Y = k) = f_k(x)$ and $\Pr(Y = k) = \pi_k$, we get

$$\Pr(X = x) = \sum_{k=1}^{K} \pi_k f_k(x).$$

Classify to the highest density



We classify a new point according to which density is highest.

When the priors are different, we take them into account as well, and compare $\pi_k f_k(x)$. On the right, we favor the pink class — the decision boundary has shifted to the left.

Why discriminant analysis?

- When the classes are well-separated, the parameter estimates for the logistic regression model are surprisingly unstable. Linear discriminant analysis does not suffer from this problem.
- If n is small and the distribution of the predictors X is approximately normal in each of the classes, the linear discriminant model is again more stable than the logistic regression model.
- Linear discriminant analysis is popular when we have more than two response classes, because it also provides low-dimensional views of the data.

Linear Discriminant Analysis when $p = 1$

The Gaussian density has the form

$$f_k(x) = \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma_k}\right)^2}$$

Here μ_k is the mean, and σ_k^2 the variance (in class k). We will assume that all the $\sigma_k = \sigma$ are the same.

Plugging this into Bayes formula, we get a rather complex expression for $p_k(x) = \Pr(Y = k|X = x)$:

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_k}{\sigma}\right)^2}}{\sum_{l=1}^{K} \pi_l \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu_l}{\sigma}\right)^2}}$$

Happily, there are simplifications and cancellations. 🤩

Discriminant functions

To classify at the value $X = x$, we need to see which of the $p_k(x)$ is largest. Taking logs, and discarding terms that do not depend on k, we see that this is equivalent to assigning x to the class with the largest *discriminant score*:

$$\delta_k(x) = x \cdot \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

Note that $\delta_k(x)$ is a *linear* function of x.

If there are $K = 2$ classes and $\pi_1 = \pi_2 = 0.5$, then one can see that the *decision boundary* is at

$$x = \frac{\mu_1 + \mu_2}{2}.$$

(See if you can show this)

Some math!

- $\begin{cases} \delta_1(x) = ? \\ \delta_2(x) = ? \end{cases}$

- Let $\delta_1(x) = \delta_2(x)$
- Show that math simplifications yields: $x = \dfrac{\mu_1 + \mu_2}{2}$

Some math! — proof

- Equal priors; the boundary is where the two scores are equal:

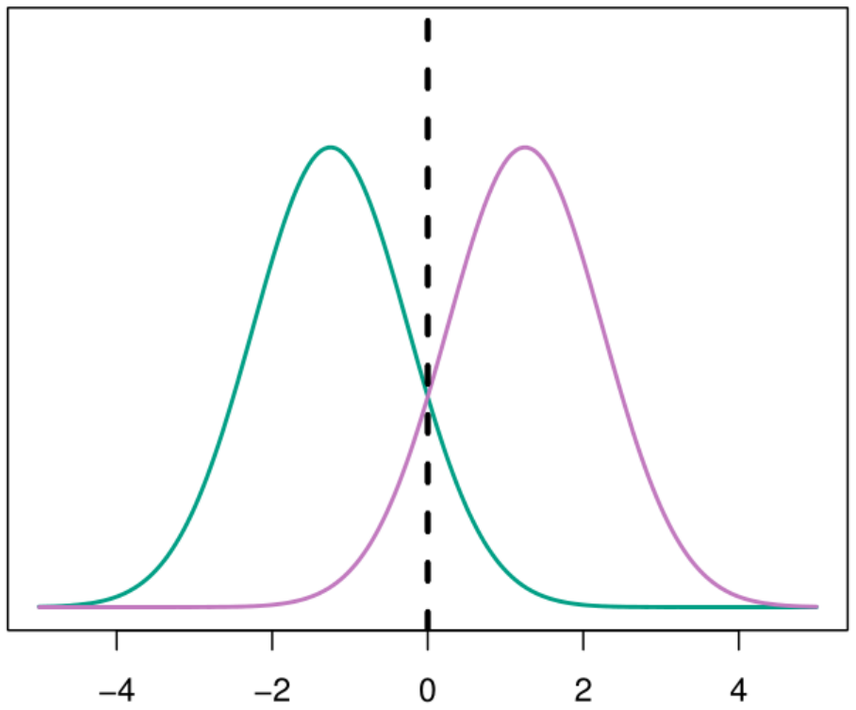
$$\delta_k(x) = x\,\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k, \qquad \pi_1 = \pi_2 = 0.5$$

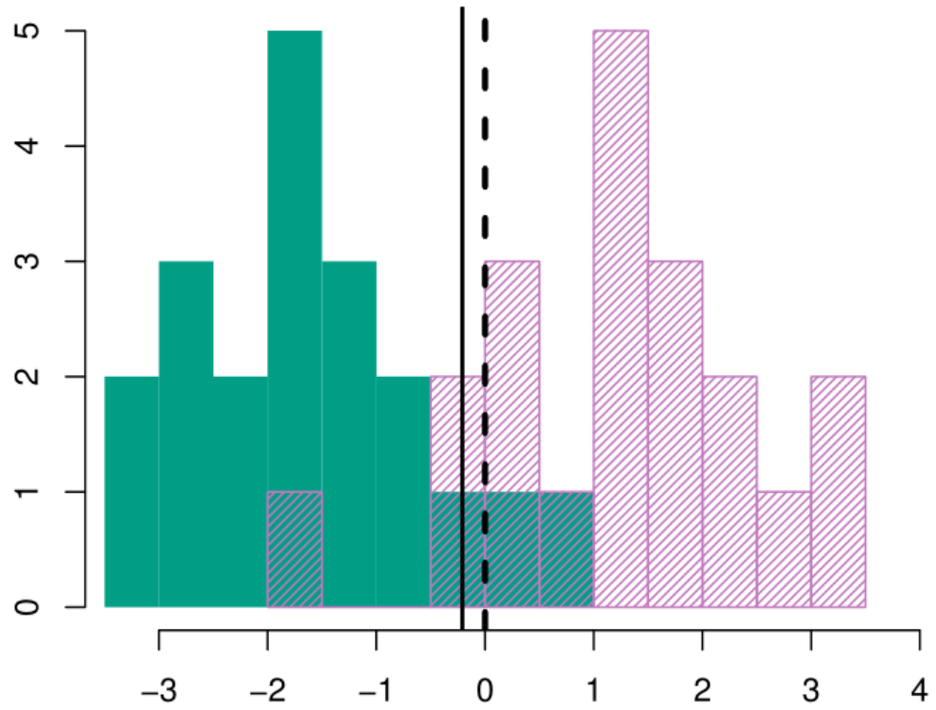
$$\delta_1(x) = \delta_2(x) \iff x\,\frac{\mu_1}{\sigma^2} - \frac{\mu_1^2}{2\sigma^2} + \log 0.5 = x\,\frac{\mu_2}{\sigma^2} - \frac{\mu_2^2}{2\sigma^2} + \log 0.5$$

$$\iff x\,\frac{\mu_1 - \mu_2}{\sigma^2} = \frac{\mu_1^2 - \mu_2^2}{2\sigma^2}$$

$$\iff x\,(\mu_1 - \mu_2) = \frac{(\mu_1 - \mu_2)(\mu_1 + \mu_2)}{2}$$

$$\iff x = \frac{\mu_1 + \mu_2}{2} \qquad (\mu_1 \neq \mu_2)$$





Example with $\mu_1 = -1.5$, $\mu_2 = 1.5$, $\pi_1 = \pi_2 = 0.5$, and $\sigma^2 = 1$.

Typically we don't know these parameters; we just have the training data. In that case we simply estimate the parameters and plug them into the rule.

Estimating the parameters

$$\begin{aligned}
\hat{\pi}_k &= \frac{n_k}{n} \\
\hat{\mu}_k &= \frac{1}{n_k} \sum_{i:\, y_i = k} x_i
\end{aligned}$$

$$\begin{aligned}
\hat{\sigma}^2 &= \frac{1}{n-K} \sum_{k=1}^{K} \sum_{i:\, y_i = k} (x_i - \hat{\mu}_k)^2 \\
&= \sum_{k=1}^{K} \frac{n_k - 1}{n-K} \cdot \hat{\sigma}_k^2
\end{aligned}$$

where $\hat{\sigma}_k^2 = \frac{1}{n_k - 1} \sum_{i:\, y_i = k} (x_i - \hat{\mu}_k)^2$ is the usual formula for the estimated variance in the kth class.

Workbook A1 — LDA scores (7 min) (ISLP §4.4.1, eq. 4.18, numbers simplified)

$X =$ months late on the last bill (simplified numbers).

Paid: $\mu_0 = 0,\ \pi_0 = 0.8$. Defaulted: $\mu_1 = 1,\ \pi_1 = 0.2$.
Pooled $\sigma^2 = 1$.

$$\delta_k(x) = x\,\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k$$

A client is one month late: $X = 1$. Compute $\delta_0(1)$ and $\delta_1(1)$. Which class does LDA pick?

Workbook A2 — the boundary (7 min) (Advanced: ISLP eq. 4.19 extended to unequal priors)

X = months late on the last bill (simplified numbers).

Paid: $\mu_0 = 0,\ \pi_0 = 0.8.$ Defaulted: $\mu_1 = 1,\ \pi_1 = 0.2.$
Pooled $\sigma^2 = 1$.

$$\delta_k(x) = x\,\frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log \pi_k$$

Solve $\delta_0(x) = \delta_1(x)$ for the boundary x^*. Compare it with the midpoint of the two means. Why is it not halfway?

Generatal decision boundary at LDA

Any priors π_1, π_2; shared variance σ^2; $\mu_1 \neq \mu_2$.

$$\delta_1(x) = \delta_2(x) \iff x\,\frac{\mu_1 - \mu_2}{\sigma^2} = \frac{\mu_1^2 - \mu_2^2}{2\sigma^2} + \log\frac{\pi_2}{\pi_1}$$

$$\iff x^* = \underbrace{\frac{\mu_1 + \mu_2}{2}}_{\text{midpoint}} + \underbrace{\frac{\sigma^2}{\mu_1 - \mu_2}\log\frac{\pi_2}{\pi_1}}_{\text{shift from the priors}}$$

- $\pi_1 = \pi_2$: $\log 1 = 0$, so x^* is the midpoint (page 53).
- $\pi_1 \neq \pi_2$: x^* moves toward the rarer class, so the more common class gets the larger region.
- The shift grows with σ^2 and shrinks as the means move apart.

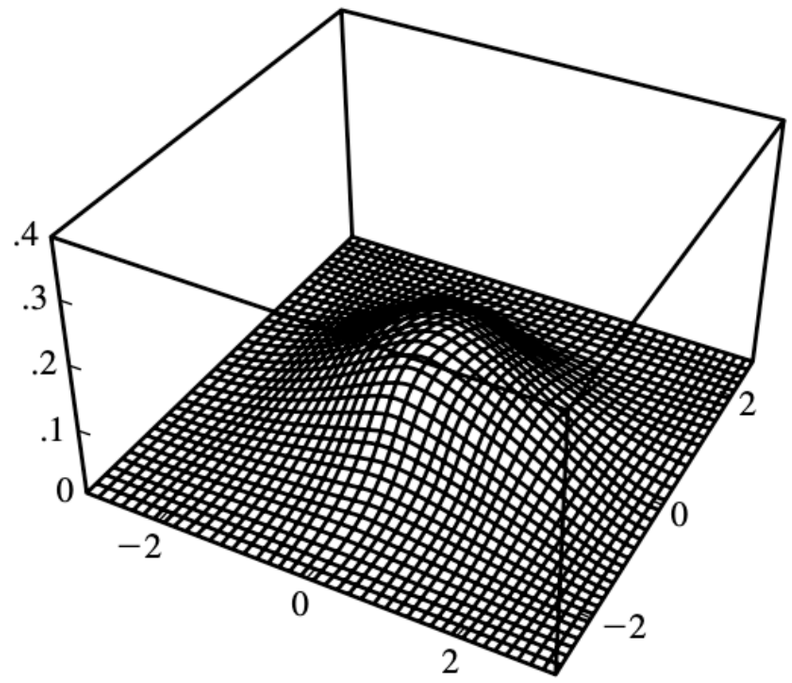
LDA when $p > 1$: Multivariate Normal Density

The bivariate normal density is

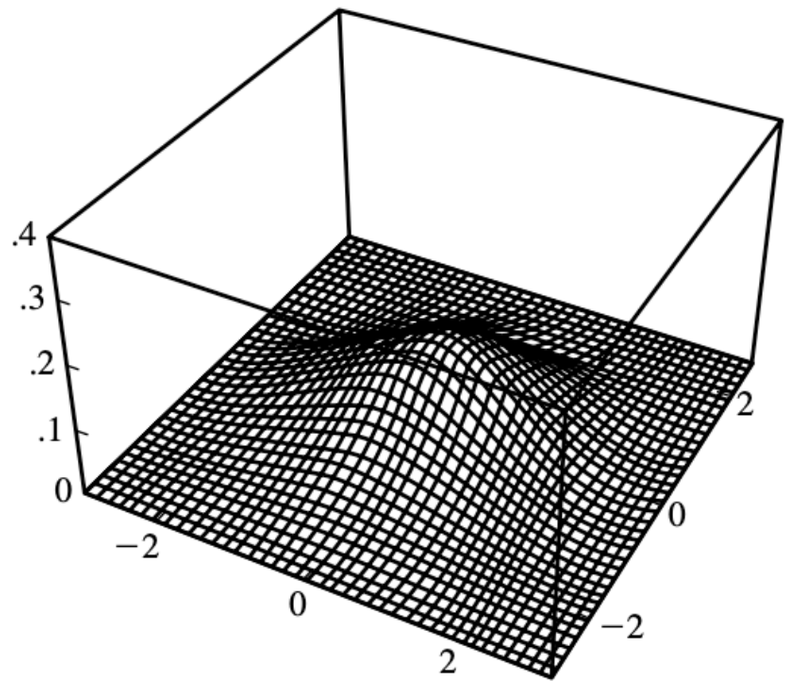
$$
\begin{aligned}
& f(x, y) \\
= \quad & \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} \right] \right),
\end{aligned}
$$

for $-\infty \leq x,\ y \leq \infty$.

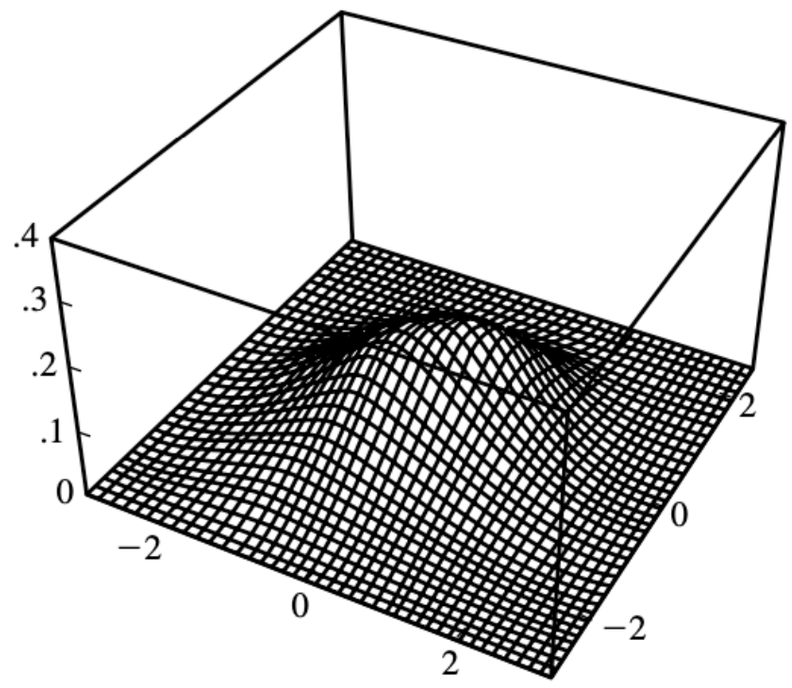
Denote $\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix} \right)$



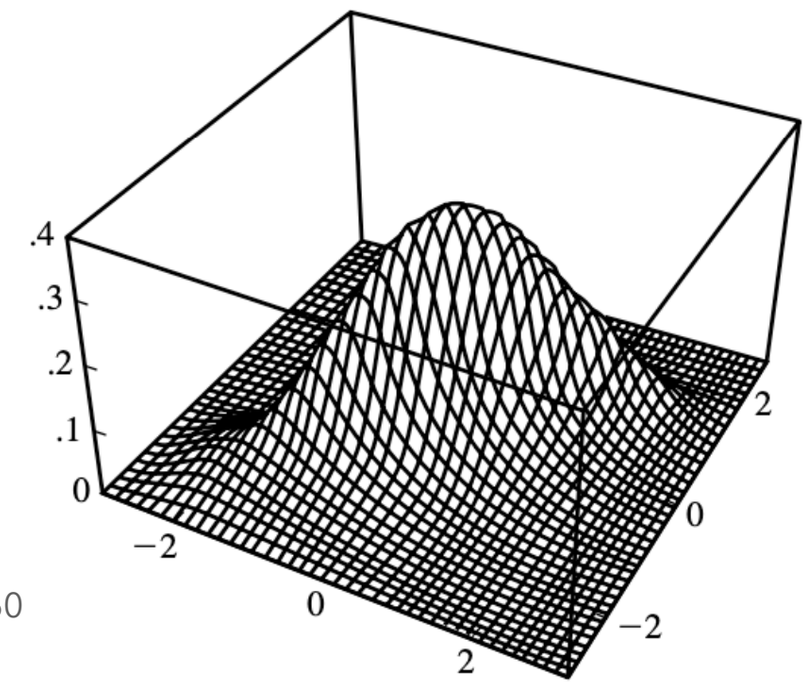
(a)



(b)



(c)



(d)

FIGURE **3.9** Bivariate normal densities with $\mu_X = \mu_Y = 0$ and $\sigma_X = \sigma_Y = 1$ and (a) $\rho = 0$, (b) $\rho = .3$, (c) $\rho = .6$, (d) $\rho = .9$.

Understand the contour lines of the density:

$$\frac{(x-\mu_X)^2}{\sigma_X^2}+\frac{(y-\mu_Y)^2}{\sigma_Y^2}-\frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y}=\text{constant.}$$

See a contour for example.

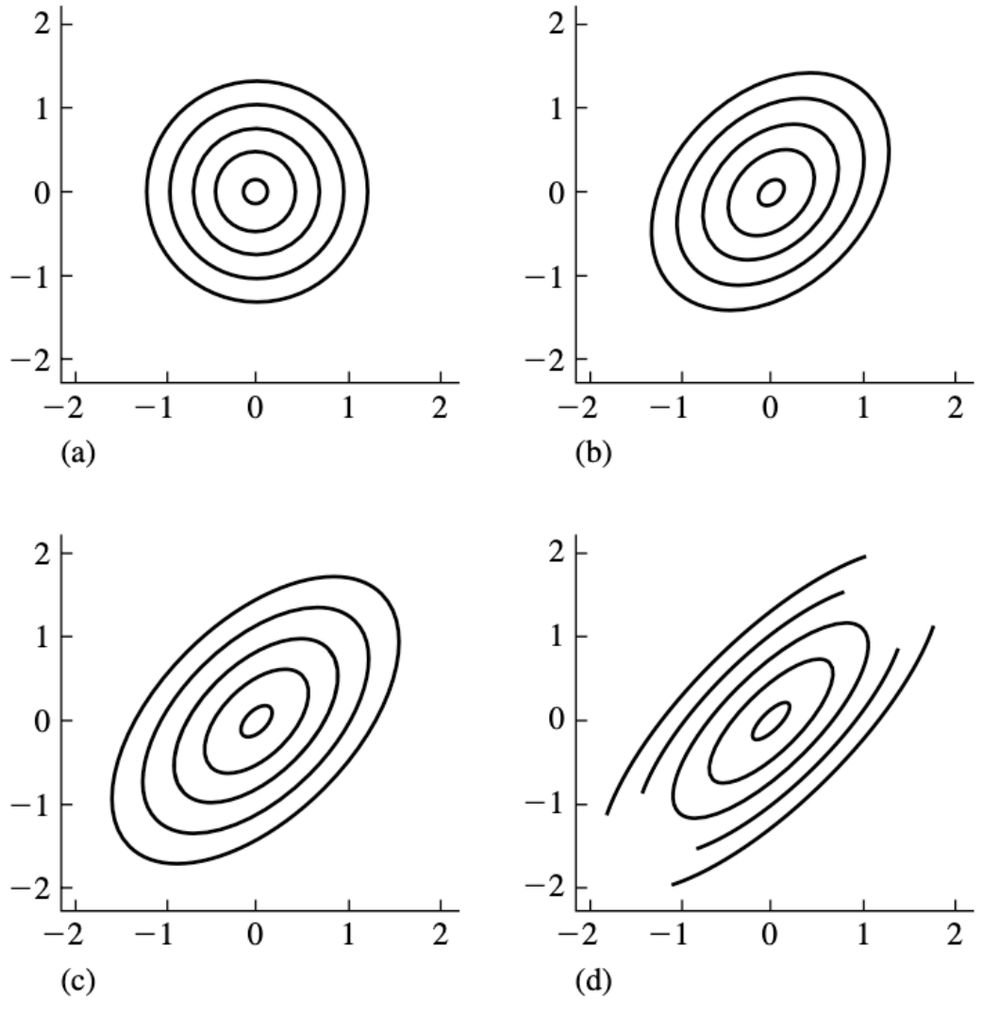


FIGURE **3.10** The elliptical contours of the bivariate normal densities of Figure 3.9.

$$X \sim N_p(\mu, \Sigma)$$

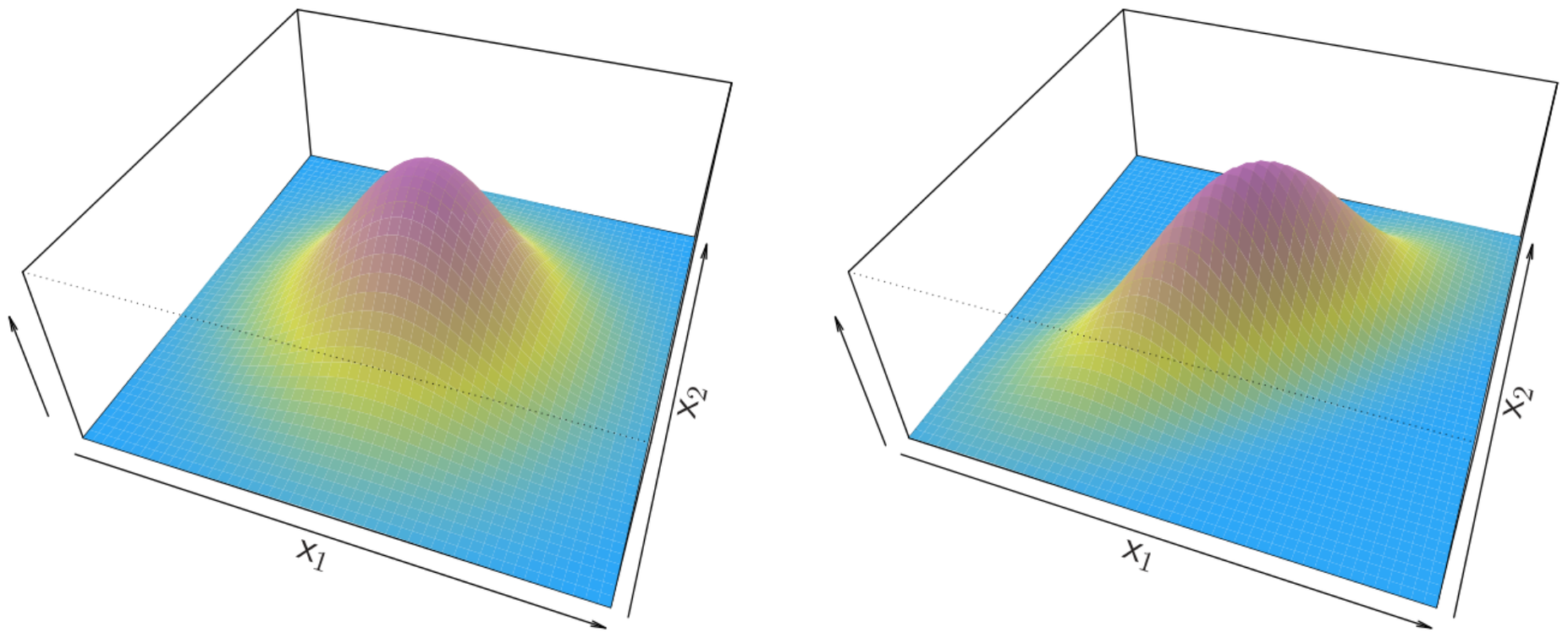
The multivariate normal density of $X \sim N_d(\mu, \Sigma)$ is

$$f(x) = \frac{1}{\sqrt{(2\pi)^p \,|\, \Sigma \,|}} \exp\left(-\frac{(x-\mu)'\Sigma^{-1}(x-\mu)}{2} \right),$$

for $-\infty \leq x_i \leq \infty,\ i = 1, \cdots, p.$

Show how the contour is affected by Σ using linear algebra!

Linear Discriminant Analysis when $p > 1$

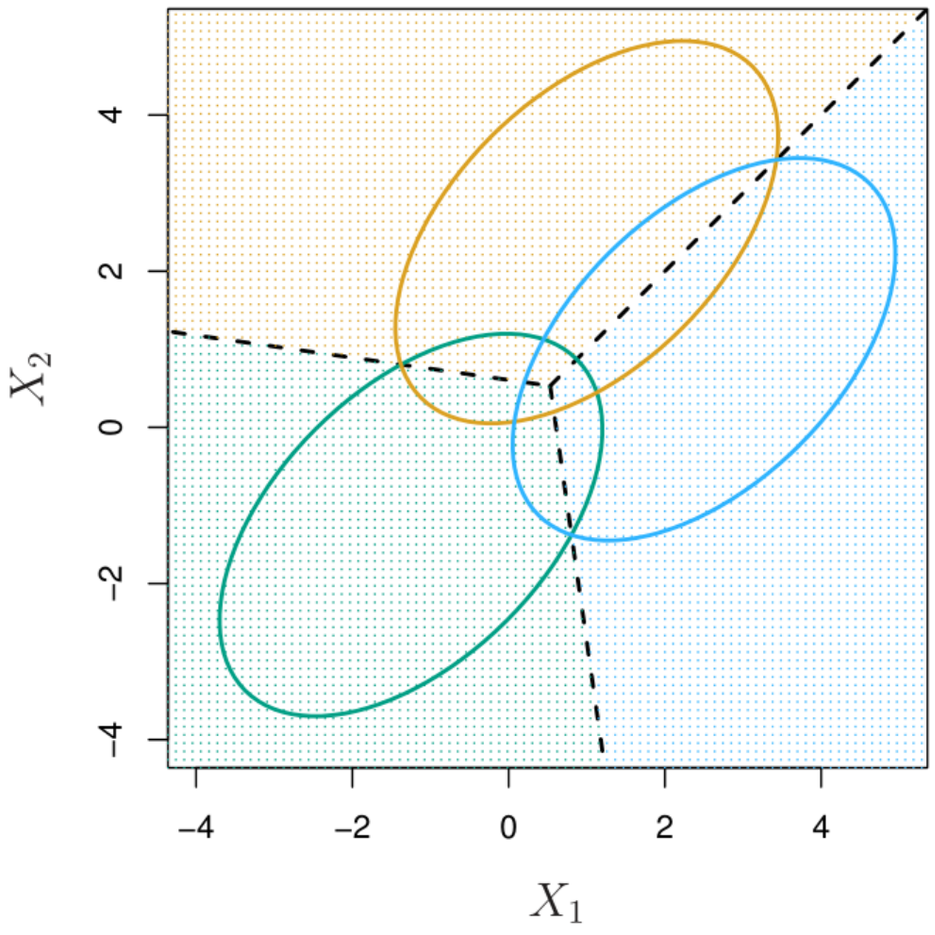


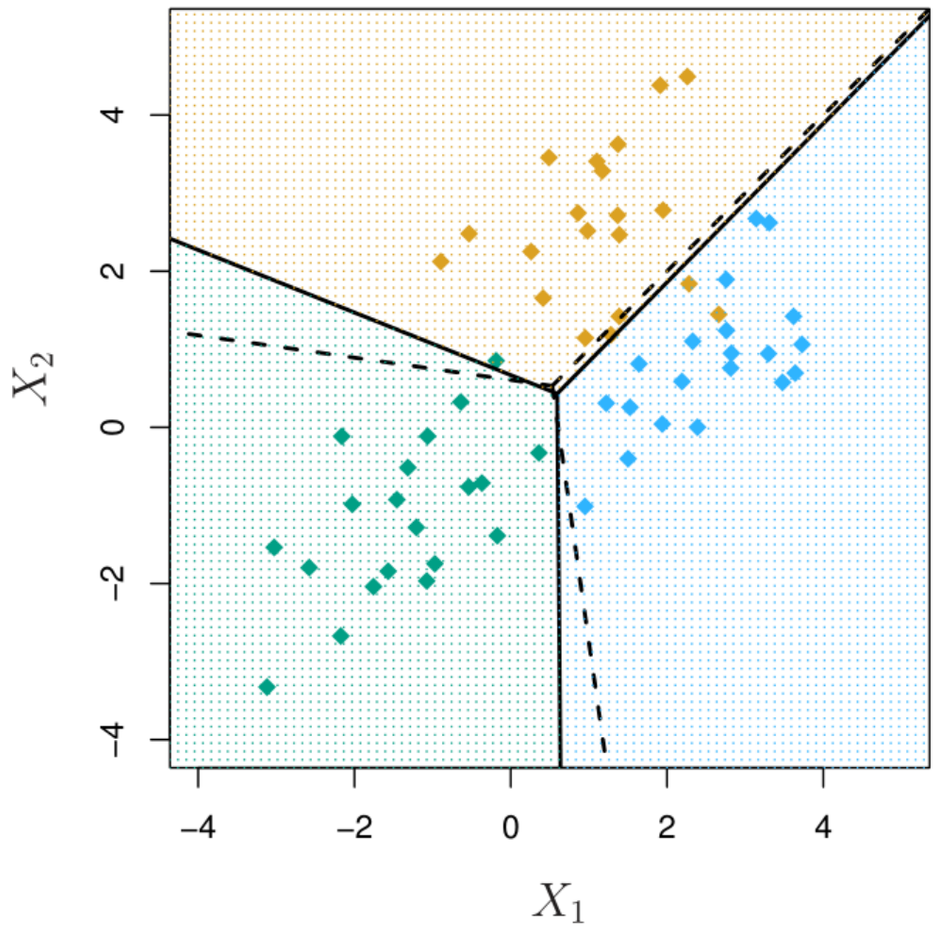
$$\text{Density: } f(x) = \frac{1}{(2\pi)^{p/2}|\mathbf{\Sigma}|^{1/2}} e^{-\frac{1}{2}(x-\mu)^T \mathbf{\Sigma}^{-1}(x-\mu)}$$

$$\text{Discriminant function: } \delta_k(x) = x^T \mathbf{\Sigma}^{-1}\mu_k - \frac{1}{2}\mu_k^T \mathbf{\Sigma}^{-1}\mu_k + \log \pi_k$$

Despite its complex form,
$\delta_k(x) = c_{k0} + c_{k1}x_1 + c_{k2}x_2 + \ldots + c_{kp}x_p$ — a linear function.

Illustration: $p = 2$ and $K = 3$ classes



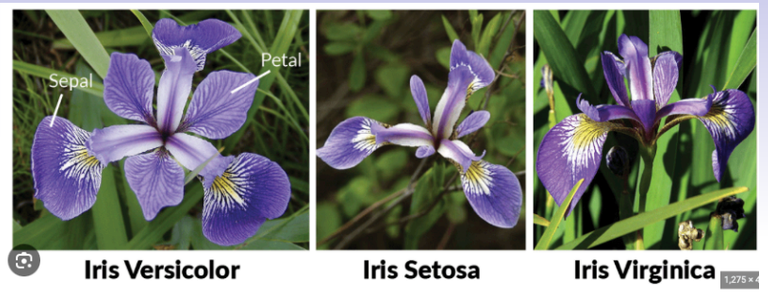


Here $\pi_1 = \pi_2 = \pi_3 = 1/3$.

The dashed lines are known as the *Bayes decision boundaries*.

Were they known, they would yield the fewest misclassification errors, among all possible classifiers.

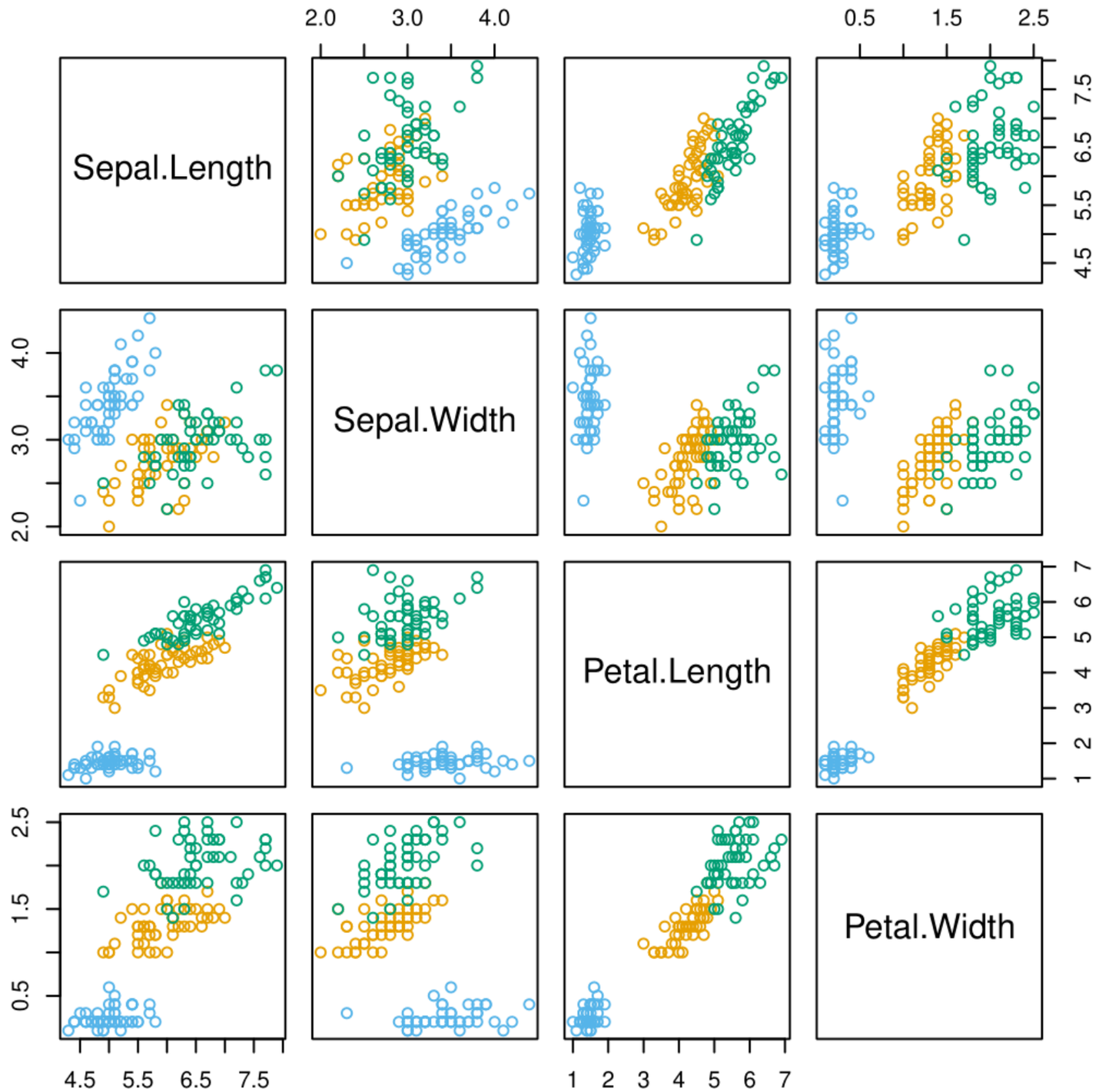
Fisher's Iris Data



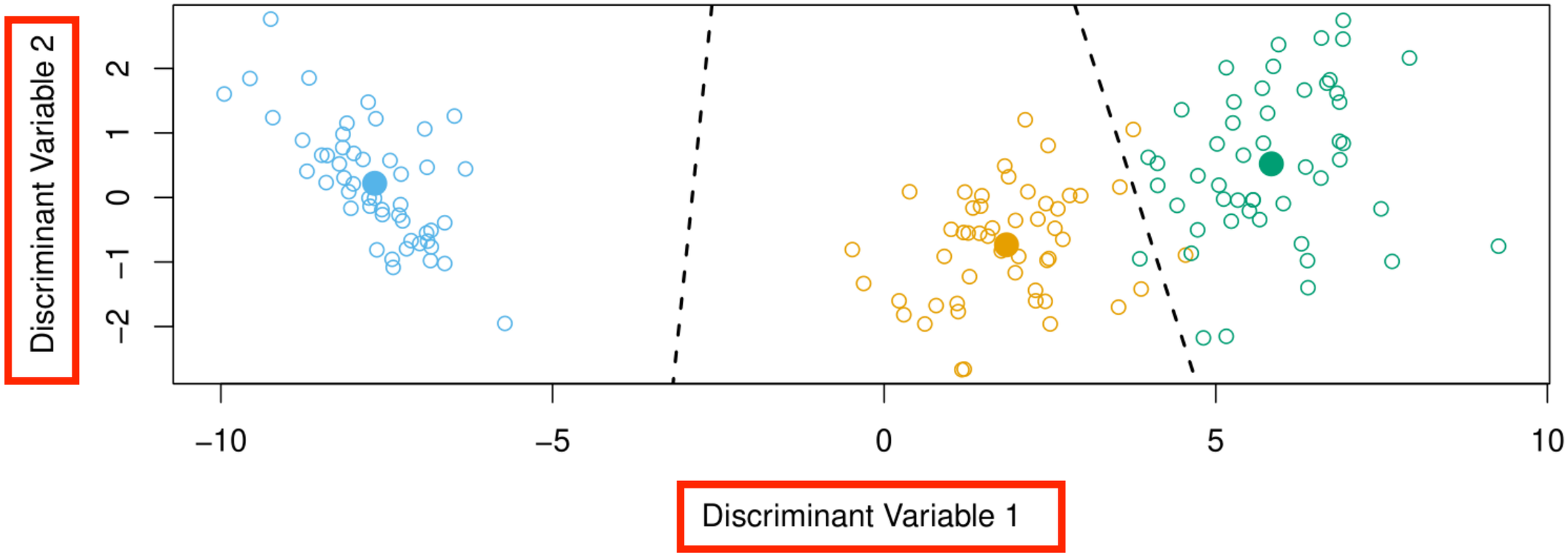
4 variables
3 species
50 samples/class

- Setosa
- Versicolor
- Virginica

LDA classifies all but 3 of the 150 training samples correctly.



Fisher's Discriminant Plot



When there are K classes, linear discriminant analysis can be viewed exactly in a $K-1$ dimensional plot.
Why? Because it essentially classifies to the closest centroid, and they span a $K-1$ dimensional plane.
Even when $K>3$, we can find the "best" 2-dimensional plane for visualizing the discriminant rule.

From $\delta_k(x)$ to probabilities

Once we have estimates $\hat{\delta}_k(x)$, we can turn these into estimates for class probabilities:

$$\widehat{\Pr}(Y = k|X = x) = \frac{e^{\hat{\delta}_k(x)}}{\sum_{l=1}^{K} e^{\hat{\delta}_l(x)}}.$$

So classifying to the largest $\hat{\delta}_k(x)$ amounts to classifying to the class for which $\widehat{\Pr}(Y = k|X = x)$ is largest.

When $K = 2$, we classify to class 2 if $\widehat{\Pr}(Y = 2|X = x) \geq 0.5$, else to class 1.

LDA on Credit Data

What is the FPR?
Waht is the FNR?

			True Default Status		
			N	P	
			No	Yes	Total
Predicted	N	No	9644	252	9896
Default Status	P	Yes	23	81	104
		Total	9667	333	10000

$(23 + 252)/10000$ errors — a 2.75% misclassification rate!

Some caveats:

- This is *training* error, and we may be overfitting. Not a big concern here since $n = 10000$ and $p = 2$!
- If we classified to the prior — always to class `No` in this case — we would make 333/10000 errors, or only 3.33%.
- Of the true `No`'s, we make $23/9667 = 0.2\%$ errors; of the true `Yes`'s, we make $252/333 = 75.7\%$ errors!

Types of errors

Check in class!

False positive rate: The fraction of negative examples that are classified as positive — 0.2% in example.

False negative rate: The fraction of positive examples that are classified as negative — 75.7% in example.

We produced this table by classifying to class `Yes` if

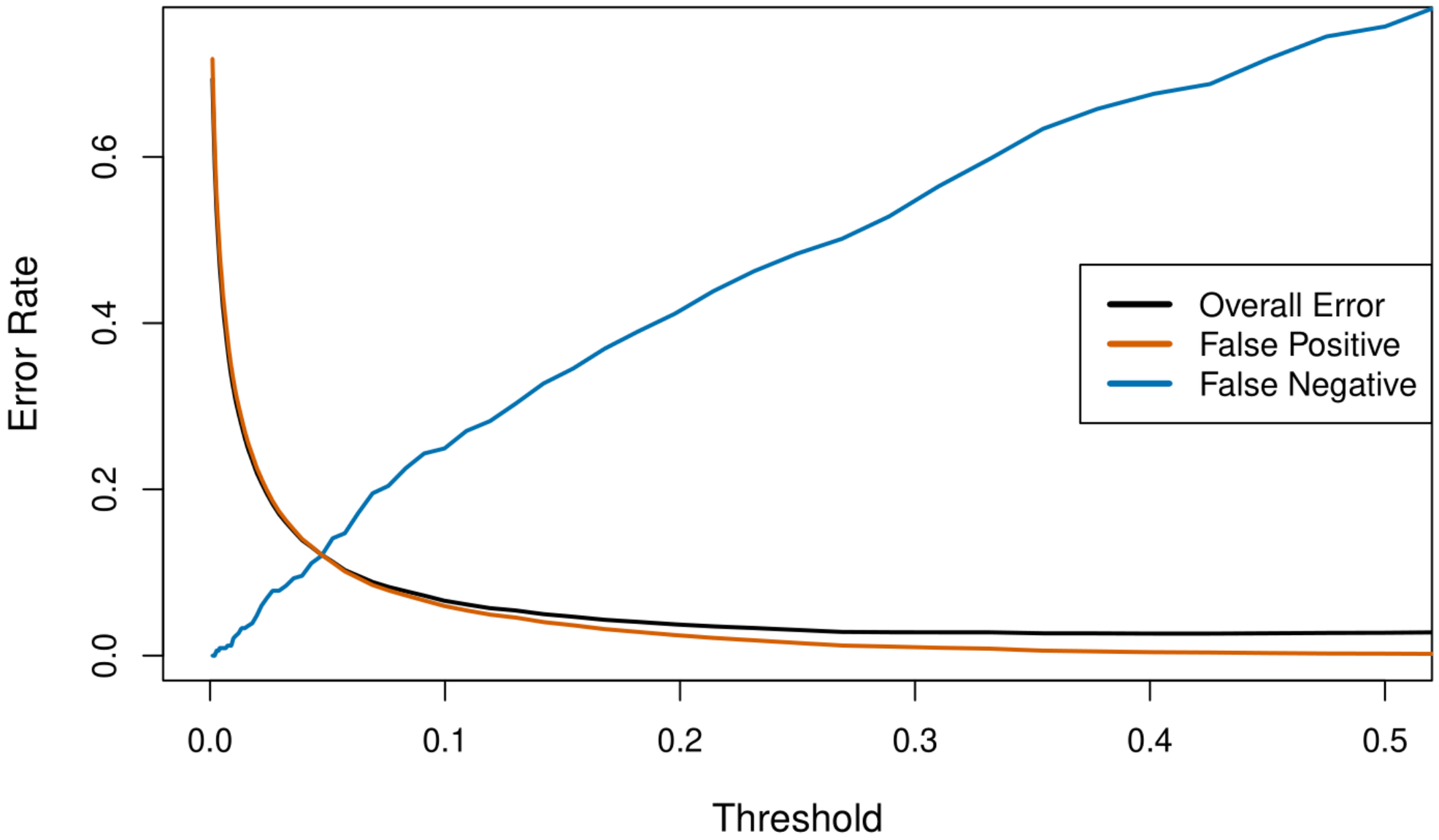
$$\widehat{\Pr}(\texttt{Default} = \texttt{Yes}|\texttt{Balance}, \texttt{Student}) \geq 0.5$$

We can change the two error rates by changing the threshold from 0.5 to some other value in $[0, 1]$:

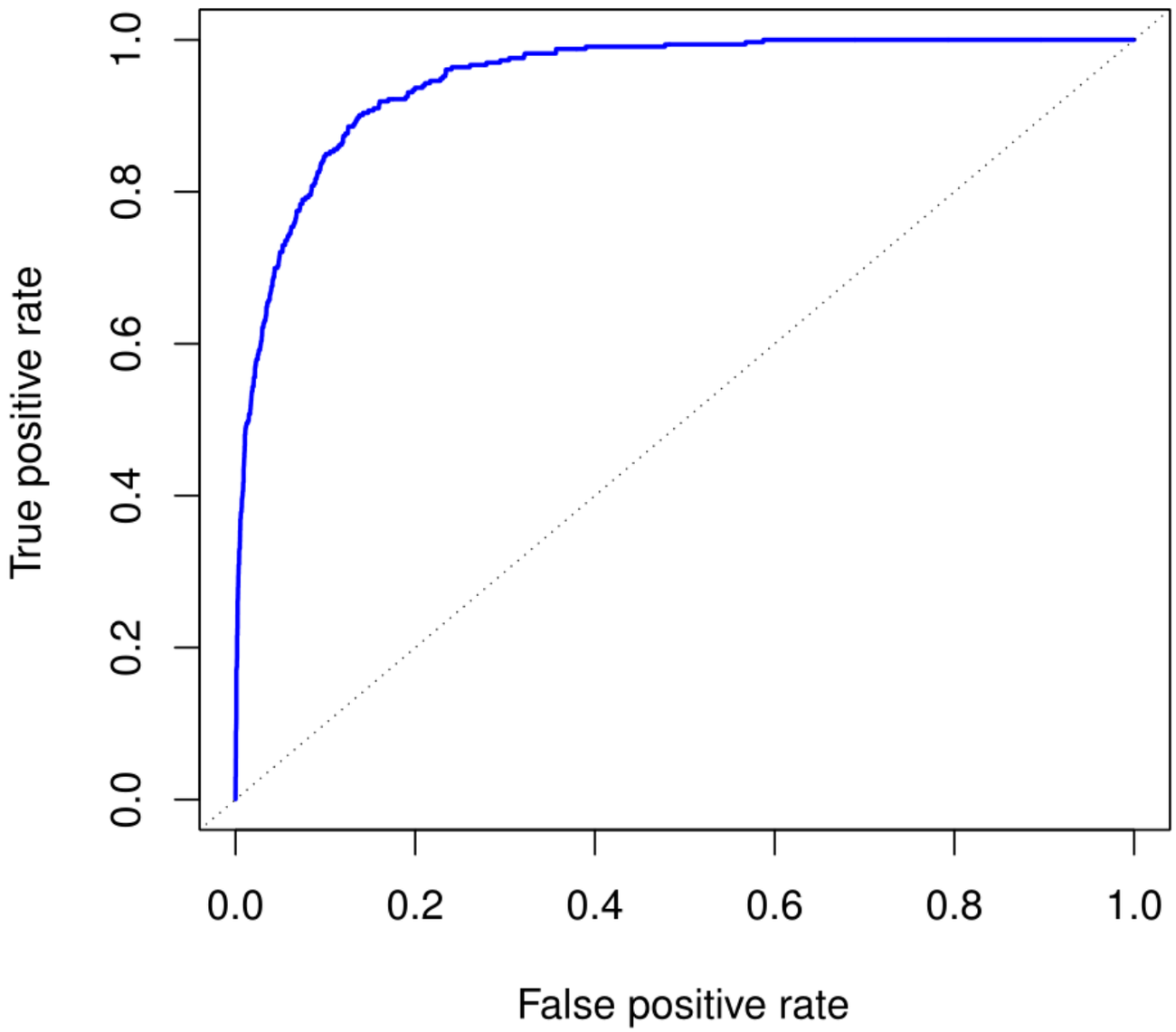
$$\widehat{\Pr}(\texttt{Default} = \texttt{Yes}|\texttt{Balance}, \texttt{Student}) \geq \textit{threshold},$$

and vary *threshold*.

Varying the *threshold*



In order to reduce the false negative rate, we may want to reduce the threshold to 0.1 or less.



The *ROC plot* displays both simultaneously.

Sometimes we use the *AUC* or *area under the curve* to summarize the overall performance. Higher *AUC* is good.

Workbook A3 — move the threshold (3 min) (ISLP §4.4.2, eq. 4.26–4.27)

		True default status		
		No	Yes	Total
Predicted default status	No	9644	252	9896
	Yes	23	81	104
	Total	9667	333	10000

TABLE 4.4. *A confusion matrix compares the LDA predictions to the true default statuses for the* 10,000 *training observations in the* `Default` *data set. Elements on the diagonal of the matrix represent individuals whose default statuses were correctly predicted, while off-diagonal elements represent individuals that were misclassified. LDA made incorrect predictions for* 23 *individuals who did not default and for* 252 *individuals who did default.*

		True default status		
		No	Yes	Total
Predicted default status	No	9432	138	9570
	Yes	235	195	430
	Total	9667	333	10000

TABLE 4.5. *A confusion matrix compares the LDA predictions to the true default statuses for the* 10,000 *training observations in the* `Default` *data set, using a modified threshold value that predicts default for any individuals whose posterior default probability exceeds* 20 %.

Use the model of A1–A2:
$\mu_0 = 0$, $\mu_1 = 1$, $\sigma^2 = 1$, $\pi_0 = 0.8$, $\pi_1 = 0.2$.

The bank fears missed defaults and predicts default when the posterior $p_1(x)$ is above $t = 0.2$, not 0.5.

Find the cut-off on x for each t.
What happens to sensitivity and to specificity?
Does the model change?

Workbook A3 — answers

A1 model: $\mu_0 = 0$, $\mu_1 = 1$, $\sigma^2 = 1$, $\pi_0 = 0.8$, $\pi_1 = 0.2$.

$$\log \frac{p_1(x)}{p_0(x)} = \delta_1(x) - \delta_0(x) = x - 0.5 + \log \frac{0.2}{0.8} = x - 1.886$$

Predict default when $p_1(x) > t$, i.e. when $x - 1.886 > \log \frac{t}{1-t}$:

$$t = 0.5: \quad x > 1.886 + \log 1 = 1.886$$

$$t = 0.2: \quad x > 1.886 + \log 0.25 = 1.886 - 1.386 = 0.5$$

	$t = 0.5$	$t = 0.2$
Sensitivity $= 1 - \Phi(c - 1)$	0.188	0.691
Specificity $= \Phi(c)$	0.970	0.691
ISLP Default data, Tables 4.4 and 4.5:		
Sensitivity	$\frac{81}{333} = 0.243$	$\frac{195}{333} = 0.586$
Specificity	$\frac{9644}{9667} = 0.998$	$\frac{9432}{9667} = 0.976$

Sensitivity rises, specificity falls. The model (μ_k, σ^2, π_k) is the same; only the cut-off c moves.

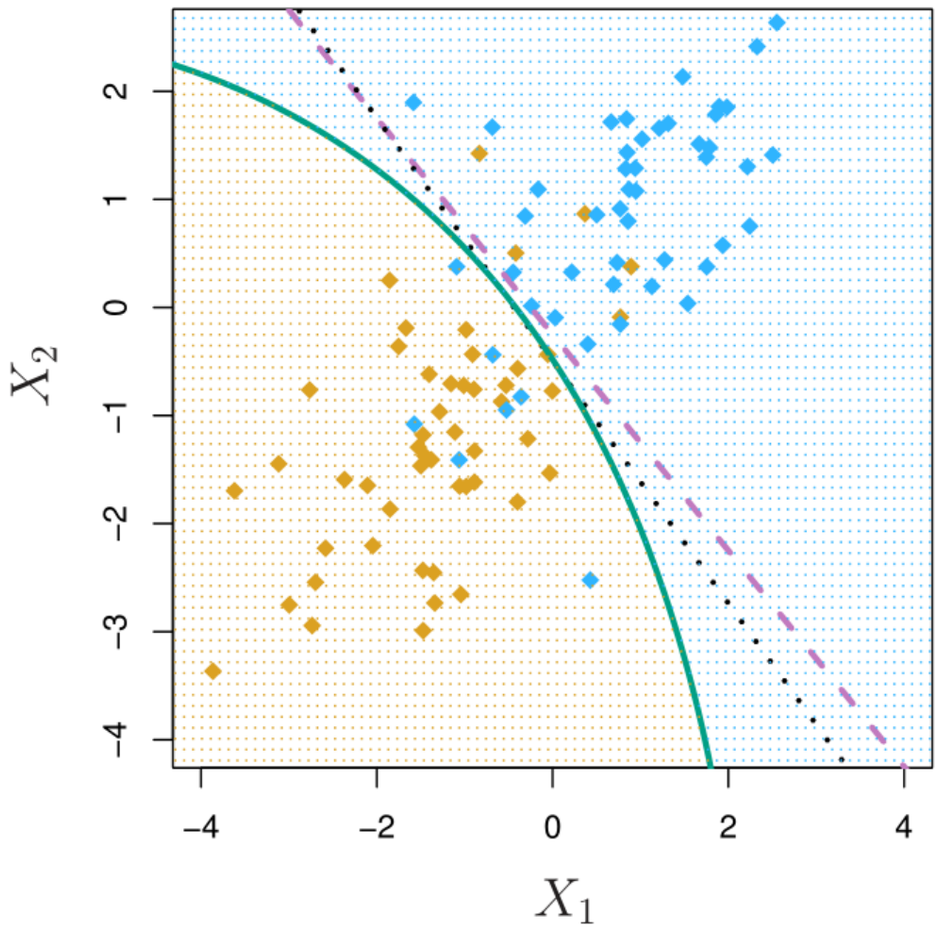
Other forms of Discriminant Analysis

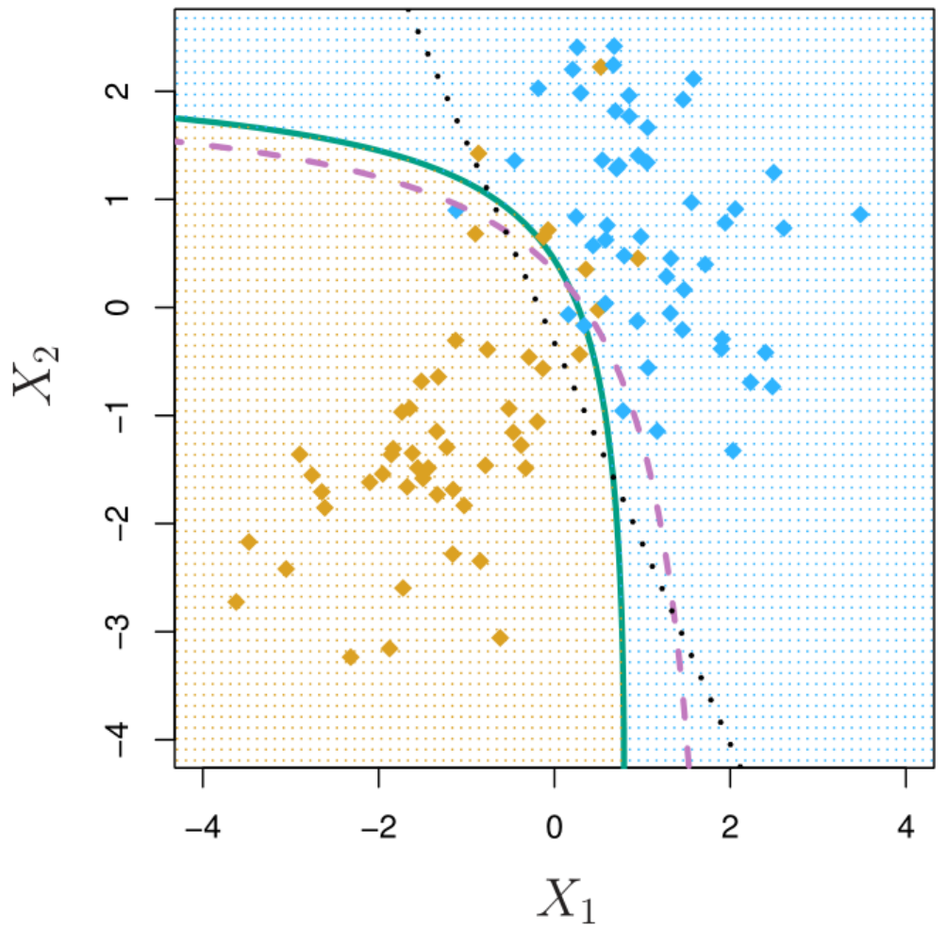
$$\Pr(Y = k|X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

When $f_k(x)$ are Gaussian densities, with the same covariance matrix $\mathbf{\Sigma}$ in each class, this leads to linear discriminant analysis. By altering the forms for $f_k(x)$, we get different classifiers.

- With Gaussians but different $\mathbf{\Sigma}_k$ in each class, we get *quadratic discriminant analysis*.
- With $f_k(x) = \prod_{j=1}^{p} f_{jk}(x_j)$ (conditional independence model) in each class we get *naive Bayes*. For Gaussian this means the $\mathbf{\Sigma}_k$ are diagonal.
- Many other forms, by proposing specific density models for $f_k(x)$, including nonparametric approaches.

Quadratic Discriminant Analysis





$$\delta_k(x) = -\frac{1}{2}(x-\mu_k)^T\mathbf{\Sigma}_k^{-1}(x-\mu_k) + \log\pi_k - \frac{1}{2}\log|\Sigma_k|$$

Because the $\mathbf{\Sigma}_k$ are different, the quadratic terms matter.

Workbook B1 — QDA scores (10 min) (ISLP §4.4.3, eq. 4.28 with one predictor)

Class 0: $\mu_0 = 0$, $\sigma_0^2 = 1$.

Class 1: $\mu_1 = 2$, $\sigma_1^2 = 4$.

Priors: $\pi_0 = \pi_1 = 0.5$.

$$\delta_k(x) = -\frac{(x-\mu_k)^2}{2\sigma_k^2} - \tfrac{1}{2}\log\sigma_k^2 + \log\pi_k$$

Compute both scores at $x = 1$ and at $x = 4$.
Which class at each point?

Would LDA, with one shared variance, have a single boundary?

Workbook B2 — count the parameters (4 min) (ISLP §4.4.3)

Two classes, p predictors.

LDA estimates one mean vector per class (p numbers each) and **one** covariance matrix shared by both classes.

QDA estimates one mean vector and **one covariance matrix per class**.

A $p \times p$ covariance matrix is symmetric, so it has $\dfrac{p(p+1)}{2}$ free numbers.

Count the mean and covariance parameters of LDA and of QDA at $p = 5$ and at $p = 50$.

Workbook B3 — which model first? (5 min) (ISLP §4.4.3; §4.8 Exercise 5)

Two classes. Number of mean and covariance parameters:

	$p = 5$	$p = 50$
LDA	25	1,375
QDA	40	2,650

LDA assumes one shared covariance: more bias, less variance. QDA gives each class its own covariance: less bias, more variance. Estimates are reliable only with many more observations than parameters.

Which model would you try first

(a) with $n = 100$ observations and $p = 50$? (b) with $n =$ 100,000 observations and $p = 5$?

Naive Bayes

Assumes features are independent in each class.
Useful when p is large, and so multivariate methods like QDA and even LDA break down.

- Gaussian naive Bayes assumes each $\mathbf{\Sigma}_k$ is diagonal:

$$\begin{aligned}\delta_k(x) &\propto \log\left[\pi_k \prod_{j=1}^{p} f_{kj}(x_j)\right] \\ &= -\frac{1}{2}\sum_{j=1}^{p}\left[\frac{(x_j-\mu_{kj})^2}{\sigma_{kj}^2} + \log\sigma_{kj}^2\right] + \log\pi_k\end{aligned}$$

- can use for *mixed* feature vectors (qualitative and quantitative). If X_j is qualitative, replace $f_{kj}(x_j)$ with probability mass function (histogram) over discrete categories.

Despite strong assumptions, naive Bayes often produces good classification results.

Logistic Regression versus LDA

For a two-class problem, one can show that for LDA

$$\log\left(\frac{p_1(x)}{1-p_1(x)}\right) = \log\left(\frac{p_1(x)}{p_2(x)}\right) = c_0 + c_1 x_1 + \ldots + c_p x_p$$

So it has the same form as logistic regression.

The difference is in how the parameters are estimated.

- Logistic regression uses the conditional likelihood based on $\Pr(Y|X)$ (known as *discriminative learning*).
- LDA uses the full likelihood based on $\Pr(X, Y)$ (known as *generative learning*).
- Despite these differences, in practice the results are often very similar.

Footnote: logistic regression can also fit quadratic boundaries like QDA, by explicitly including quadratic terms in the model.

Summary

- Logistic regression is very popular for classification, especially when $K = 2$.
- LDA is useful when n is small, or the classes are well separated, and Gaussian assumptions are reasonable. Also when $K > 2$.
- Naive Bayes is useful when p is very large.
- See Section 4.5 for some comparisons of logistic regression, LDA and KNN.